# Research Log Book

**Ronak Patel**

Project: Knowledge Tracing for Large Language Models

CMPE 295A, San José State University, Fall 2026 · Advisor: Dr. Mahima Suresh Agumbe

Team: Ronak Patel, Bhagyesh Rathi, Aniket Sarrin · Repository: github.com/bhagyeshrathi07/LLM-Knowledge-Tracing

Period covered: August 31 – October 29, 2026 · Prepared October 29, 2026

## Role and track

Data and infrastructure, evaluation harness, visualization (track owner since Sep 29); learner-model experiments (T-SKIRT, recalibration, calibration, DriftIRT); project notes and meeting notes.

## Summary of contribution to date

- Wrote the abstract's method paragraph and the abstract v2 rewrite; ran the systematic literature search and commissioned the deep review.
- Verified the Fluid Benchmarking release and built the whole Phase 1 pipeline: ingest (12.6 million responses), per-skill curves, walk-forward harness, metrics, six baselines, chronological validation/test blocks, bootstrap standard errors, paired differences.
- Built and scored T-SKIRT, per-model recalibration, walk-forward calibration and DriftIRT; built the baseline-ladder analysis; audited every result that beat the bar.
- Repository: 16 commits on main as of Oct 28 (about 7,700 lines added), five pull requests (#1 to #4 opened, #1 to #3 merged); test suite grew from 50 to 412 tests across the team.
- Assembled Workbook 1 and its corrections, the meeting-notes folder, the project notes (Decisions 1 to 6), two team briefings, the advisor briefing of Oct 19, and the presentation speaker notes (slides 7 to 12 delivered).

## Weekly log

Each entry lists the meetings attended, the work done that week by this member (as opposed to the team), and the artifacts that evidence it (repository paths, commits, documents).

## Week 1: Aug 31 – Sep 6

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| **Meetings** | Aug 31 advisor meeting (Google Meet, 11:00–11:20; note-taker). Sep 1 impromptu advisor meeting (in person, 9:30–9:45). |
| **Work done** | • Discussed the advisor's candidate project ideas; the team chose knowledge tracing for LLMs across training checkpoints and refined the framing with her (checkpoint = learner state, skill-tagged items = evidence).<br>• Sep 1: discussed with the advisor what 'skill' and 'time step' mean for an LLM, what evidence a benchmark response gives, and how the result would be validated; asked for reading material.<br>• Sep 2: drafted abstract paragraph 3 (method, data, evaluation); merged the three paragraphs into one abstract and e-mailed it to the advisor.<br>• Sep 3: submitted the abstract on Canvas (assignment due Sep 8); received the advisor's e-mail feedback the same day (NeurIPS target, add a visualization component, simplify the last paragraph) and recorded it in the meeting notes.<br>• Drafted the repository structure (src/, scripts/, tests/, data/, docs/) that the team adopted. |
| **Artifacts** | meetings/2026-08-31-advisor.md, meetings/2026-09-01-advisor.md, meetings/2026-09-03-advisor.md; Project_Abstract_KT_for_LLMs.pdf (Canvas). |

## Week 2: Sep 7 – Sep 13

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| **Meetings** | Sep 8 team meeting (Discord, 5:00–6:00). |
| **Work done** | • Walked through the advisor's feedback with the team and agreed what each point changes: NeurIPS moves the paper deadline to May 2027; visualization becomes a semester-2 deliverable; the abstract rewrite waits for the literature search.<br>• Took the deep-KT and KT-to-IRT reading: Piech et al. 2015 (DKT), Ghosh et al. 2020 (AKT), Vie & Kashima 2023 (DKT as implicit dynamic IRT). Wrote one-paragraph summaries (what it does, data, assumptions, relevance) for the shared document.<br>• Conclusion from the reading, later adopted as Decision 2: deep KT needs thousands of learner sequences, so with six models it can only be an ablation; the IRT link is the bridge our temporal models use. |
| **Artifacts** | meetings/2026-09-08-team.md; paper summaries (later folded into docs/project-notes.md §4). |

### Week 3: Sep 14 – Sep 20

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| **Meetings** | Sep 15 team meeting (Discord, 5:00–6:00). |
| **Work done** | • Presented the deep-KT summaries. Took part in the first discussion of the central design question: benchmark items are observations, not practice, so the time step should be the checkpoint with all items at a checkpoint as simultaneous noisy readings (Decision 1).<br>• Agreed the tentative two-phase plan (Phase 1 on public checkpoint data with MMLU subjects as skills; Phase 2 on OLMo-2 with MathCAMPS).<br>• Started the systematic literature search for prior work that fits a learner model to LLM checkpoints (action item, due Sep 22): query design across arXiv, Semantic Scholar, ACL Anthology and OpenReview. |
| **Artifacts** | meetings/2026-09-15-team.md. |

### Week 4: Sep 21 – Sep 27

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| **Meetings** | Sep 23 team meeting (Discord, 10:00–11:00; note-taker). |
| **Work done** | • Sep 22: completed the systematic literature search (about 25 queries). Identified and read the closest prior work: MathCAMPS (COLM 2025), Fluid Benchmarking (COLM 2025), the Implicit Curriculum paper (Apr 2026), Mapping Post-Training Forgetting (Oct 2025), Item Response Scaling Laws (Jun 2026), Hu et al. (TMLR 2023).<br>• Reported the result at the Sep 23 meeting: no prior work fits a sequential latent learner model to an LLM's checkpoint responses or validates it by predicting later checkpoints, so the core claim holds; but 'goes unmeasured' in the abstract is too strong and must become 'described but not modeled'.<br>• Confirmed which six models the abstract's 'six open models' refers to (Fluid's release). Proposed verifying the Fluid dataset by downloading it and inspecting the files instead of trusting the README; took that action item (due Sep 29) together with drafting abstract v2. |
| **Artifacts** | meetings/2026-09-23-team.md; docs/project-notes.md §4 (prior work, ranked by overlap). |

## Week 5: Sep 28 – Oct 4

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| **Meetings** | Sep 29 team meeting (Discord, 5:30–6:30). Workbook 1 submitted Oct 1. |
| **Work done** | • Sep 29: verified the data. Downloaded allenai/fluid-benchmarking with the HuggingFace client and inspected every per-model CSV: six models, 452 checkpoints, 28,659 items (27,842 after dropping TruthfulQA, whose scores are continuous), one 0/1 cell per (checkpoint, item), no missing values, plus fitted 2PL item parameters. Phase 1 needs no GPU.<br>• Set up the Python 3.12 virtual environment and pyproject packaging; wrote src/llm_kt/ingest.py (checkpoint-label and item-id parsing, wide-to-long conversion, validation) and scripts/ingest_fluid.py producing data/processed/fluid_responses.parquet (12,584,584 rows, 47 s).<br>• Wrote src/llm_kt/curves.py (per-skill accuracy with Wilson intervals) and scripts/plot_skill_curves.py; produced the OLMo-2 MMLU curve figure and the six-model biology figure; identified the three curve shapes and seven peak-to-final drop candidates.<br>• Wrote src/llm_kt/metrics.py (log-loss, Brier, accuracy, rank AUC), src/llm_kt/evaluate.py (walk-forward Predictor interface, rolling origins, probability checks) and src/llm_kt/baselines.py (global mean, skill mean, item mean, copy-last, copy-last-calibrated); ran scripts/run_baselines.py at horizons 1 and 5 on all six models and set the pre-committed bar (item mean, log-loss 0.360 / AUC 0.80 at h=1). 50 tests, ruff clean.<br>• Wrote docs/project-notes.md (plain-language project description, data verification, prior work, Decisions 1 to 5 with pros and cons, roadmap, risks) and docs/abstract-v2.md; presented Decisions 3 to 5 and the baseline results at the Sep 29 meeting, where Decisions 1 to 5 were adopted and track owners agreed.<br>• Sep 30: wrote docs/research-brief.md and commissioned the five-priority deep literature review (docs/lit-review-2026-09-30.md, about 60 queries, 45 papers read in full); recorded its consequences (dynIRT not a drop-in, BKT needs log-space and per-item emissions, drift likely undetectable, statistics plan).<br>• Sep 30 – Oct 1: assembled Workbook 1 (docs/workbook/build/content.py, HTML and DOCX renderers, architecture figure, 49 references); commissioned two independent reviews and corrected every error they found (empirical numbers, eight bibliography records, feasibility claims); wrote the eight meeting notes for Aug 31 to Sep 29 from the team's records; wrote the first team briefing (docs/team-briefing.md).<br>• Sep 30: first commits to the shared repository (code; README; meeting notes). |
| **Artifacts** | Commits 5a4abcd (ingest, curves, harness, baselines), a771145 (README), b38d094 (meeting notes); docs/workbook/Workbook1.pdf; docs/project-notes.md; docs/abstract-v2.md; docs/research-brief.md; docs/lit-review-2026-09-30.md; docs/figures/skill_curves_olmo2-7b.png, skill_high_school_biology_models.png; data/processed/baseline_results_h{1,5}.parquet. |

## Week 6: Oct 5 – Oct 11

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| **Meetings** | Oct 9: Workbook 1 presentation delivered in class (Bhagyesh slides 1–6, Ronak slides 7–12). |
| **Work done** | • Sprint 3 harness work: chronological train/validation/test checkpoint blocks per model (OLMo-2: 0–55 / 56–74 / 75–93), subject-clustered bootstrap standard errors in summarize(), and paired_difference() with 95% intervals between any two predictors (src/llm_kt/evaluate.py, tests/test_blocks.py).<br>• Wrote src/llm_kt/tskirt.py (temporal IRT with drift-discounted discrimination, probit link, posterior variance) on top of the 2PL item-parameter loader; scored it on OLMo-2 MMLU: T-SKIRT with Fluid's published item parameters loses to the bar (pooled log-loss 0.561 vs 0.457).<br>• Wrote the speaker notes for all twelve Workbook 1 presentation slides (docs/presentation-script.md, about 1,790 words, 12 minutes), plain-language versions for a general audience; delivered slides 7 to 12 on Oct 9. |
| **Artifacts** | src/llm_kt/evaluate.py, tskirt.py; tests/test_blocks.py, test_tskirt.py; scripts/run_models.py; docs/presentation-script.md; Workbook1_Presentation.pptx (speaker notes). Committed Oct 19 as 7f54a62. |

## Week 7: Oct 12 – Oct 18

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| **Meetings** | No team meeting recorded. |
| **Work done** | • Diagnosed why T-SKIRT lost: Fluid's item parameters were fitted on 102 finished models and do not describe one model's trajectory. Wrote src/llm_kt/recalibrate.py: per-model item difficulty offsets and log-discrimination multipliers fitted by Newton MAP on the model's own history with shrinkage. With recalibration T-SKIRT ties the bar on the test block (0.375 ± 0.024 vs 0.375 ± 0.015) and leads pooled over the run.<br>• Ran the hyperparameter sweeps (tau, tau_a, nu) selected on validation only (scripts/analyze_sweep.py) and the horizon study at h = 5, 10, 20 (scripts/analyze_horizons.py): T-SKIRT wins early in training, ties at short range on the test block, loses at h = 20 (+0.024 log-loss).<br>• Wrote src/llm_kt/calibrate.py (walk-forward temperature and Platt calibration) and scripts/run_calibrated.py: calibration fixes static IRT's under-confidence (0.393 to 0.382) and does nothing for T-SKIRT. Conclusion recorded: the remaining gap is ranking within a skill, not confidence.<br>• Verified every reported number against the result files before recording it. |
| **Artifacts** | src/llm_kt/recalibrate.py, calibrate.py; tests/test_recalibrate.py, test_calibrate.py; scripts/run_calibrated.py, analyze_sweep.py, analyze_horizons.py; data/processed/models_mmlu_*.parquet. 328 tests passing at the end of the sprint. |

## Week 8: Oct 19 – Oct 25

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| **Meetings** | Oct 19 advisor meeting (11:20–11:40 PST; note-taker). |
| **Work done** | • Oct 19: wrote the screen-shared advisor briefing (docs/advisor-briefing-2026-10-19.html): project, prior work, data, pipeline, results with every number re-verified from result files and labelled by metric, honest verdict, proposed next model. Presented it; recorded the advisor's interest in the baseline ladder (global 0.69 / subject 0.63 / item 0.46 log-loss), her held-out-test suggestion, and the chain-of-thought direction.<br>• Oct 19: committed and pushed Sprint 3 code and the Workbook 1 document package; created ronak_branch; wrote the Oct 19 meeting note and rewrote docs/team-briefing.md (repo map, verified results, meeting outcomes, per-person next steps); opened and merged PRs #1 and #2.<br>• Oct 20: two experiments, each verified against its result files: (a) baseline-ladder analysis (src/llm_kt/ladder.py, scripts/analyze_ladder.py, horizons 10 and 20): the item gap grows over training (OLMo-2 0.07 to 0.24) and subject explains 13% of correctness variance vs 58% for the item's own past; (b) DriftIRT (src/llm_kt/drift_irt.py): per-skill ability random walk plus per-item residual random walk, selected on validation, beats the bar on the test block at h = 1, 5, 20 (−0.069, −0.067, −0.066 log-loss).<br>• Verification work: reproduced harness rows from fresh fits at three origins, shuffled-label control (log-loss 0.31 to 1.4), and an exponentially weighted item-mean control showing that recency explains most of DriftIRT's gain. |
| **Artifacts** | meetings/2026-10-19-advisor.md; docs/team-briefing.md; commits 7f54a62, 1b032cf, a3577f0, 0469083; PRs #1, #2; src/llm_kt/ladder.py, drift_irt.py; tests/test_ladder.py, test_drift_irt.py; docs/figures/ladder_over_training.png, variance_decomposition.png. |

**Week 9: Oct 26 – Oct 29**

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| **Meetings** | No meeting. Workbook 2 and log book due Oct 29. |
| **Work done** | • Oct 27: committed the ladder analysis and DriftIRT and opened PR #3 (merged); recorded Decision 6 (learner models on MMLU only; other benchmarks kept for the ladder, held-out transfer tests and controls) in docs/project-notes.md and opened PR #4.<br>• Oct 28: pulled Bhagyesh's BKT work (15 commits), merged main into ronak_branch, reviewed the design against the pasted BKT rationale, and reproduced his headline run on my machine (two BKT settings, item mean, recency-weighted item mean at h = 1 on OLMo-2 MMLU, about 14 minutes CPU): log-loss matched his stored tables to six decimal places (test block 0.3095 / 0.3163 / 0.3747 / 0.3165).<br>• Oct 28–29: wrote Workbook 2 (chapters 6 to 9 completed, other chapters updated, Gantt and PERT charts for both semesters) and this log book; reconciled the submitted Workbook 1 text with the repository source. |
| **Artifacts** | Commits 4f5ff65, ec8706c, 6e91e5d, 10008cd, 2784c52; PRs #3, #4; data/processed/ models_bkt_cmp{S,F}_h1.parquet; docs/workbook/Workbook2.pdf; docs/logbook/. |

Signed: _____ Date: _____